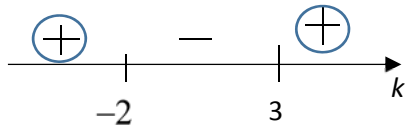
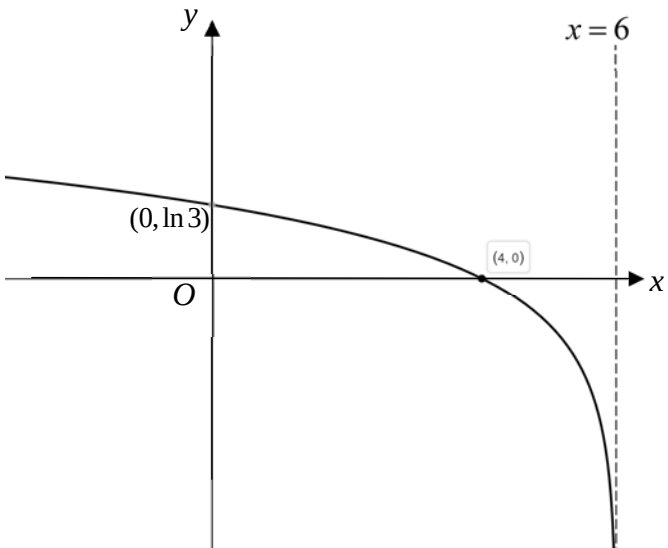
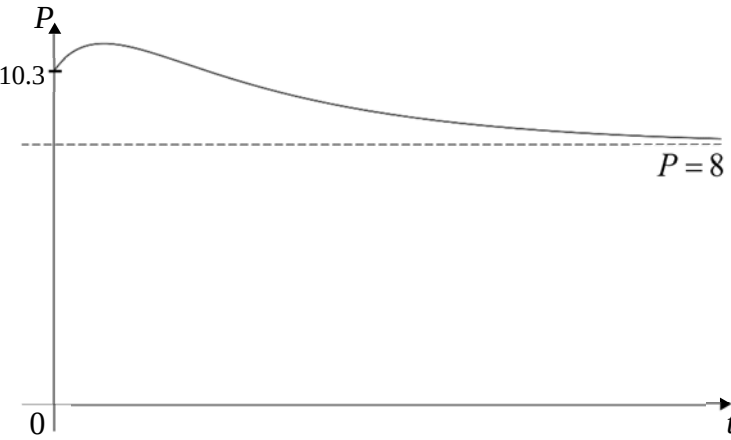


2018 IJC H1 Math Prelim Solution

Qn	Solution
1	$(k-2)x^2 - 2kx + (2k+3) < 0$ For no real roots, $b^2 - 4ac < 0$ $(-2k)^2 - 4(k-2)(2k+3) < 0$ $4k^2 - (8k^2 - 4k - 24) < 0$ $-4k^2 + 4k + 24 < 0$ $k^2 - k - 6 > 0$ $(k-3)(k+2) > 0$  $\therefore k < -2$ or $k > 3$ Since $(k-2)x^2 - 2kx + (2k+3) < 0$, coeff of x^2 must be negative, thus $k-2 < 0 \Rightarrow k < 2$. $\therefore \{k : k \in \mathbb{R}, k < -2\}$
2(a)	$\frac{d}{dx} \ln(\sqrt{1+3x^2}) = \frac{d}{dx} \left[\frac{1}{2} \ln(1+3x^2) \right]$ $= \frac{1}{2} \left(\frac{1}{1+3x^2} \right) (6x)$ $= \frac{3x}{1+3x^2}$
(b)	$\int \frac{(2\sqrt{x}-1)^2}{\sqrt{x}} dx = \int \frac{4x - 4\sqrt{x} + 1}{\sqrt{x}} dx$ $= \int 4\sqrt{x} - 4 + \frac{1}{\sqrt{x}} dx$ $= \frac{4x^{3/2}}{3/2} - 4x + \frac{x^{1/2}}{1/2} + c$ $= \frac{8}{3}x^{3/2} - 4x + 2x^{1/2} + c$

3(i)	Let r cm be the radius of the semicircles. Perimeter of the banner is $2\pi r + 4x = 12$ $r = \frac{12 - 4x}{2\pi} = \frac{6 - 2x}{\pi}$ Area of the banner is $A = 2x \left[2 \left(\frac{6 - 2x}{\pi} \right) \right] - \pi \left(\frac{6 - 2x}{\pi} \right)^2$ $= \left(\frac{6 - 2x}{\pi} \right) \left[4x - \pi \left(\frac{6 - 2x}{\pi} \right) \right]$ $= \left(\frac{6 - 2x}{\pi} \right) [6x - 6]$ $= \frac{12}{\pi} (3 - x)(x - 1)$												
(ii)	$A = \frac{12}{\pi} (3 - x)(x - 1)$ $= \frac{12}{\pi} (-x^2 + 4x - 3)$ $\frac{dA}{dx} = \frac{12}{\pi} (-2x + 4)$ For stationary values, $\frac{dA}{dx} = 0$ $\frac{dA}{dx} = \frac{12}{\pi} (-2x + 4) = 0$ $x = 2$ <table><tr><td>x</td><td>2^-</td><td>2</td><td>2^+</td></tr><tr><td>Sign of $\frac{dA}{dx}$</td><td>$+$</td><td>0</td><td>$-$</td></tr><tr><td>slope</td><td></td><td></td><td></td></tr></table> Maximum Area = $\frac{12}{\pi} \text{ m}^2$	x	2^-	2	2^+	Sign of $\frac{dA}{dx}$	$+$	0	$-$	slope			
x	2^-	2	2^+										
Sign of $\frac{dA}{dx}$	$+$	0	$-$										
slope													

4(i)	Vertical asymptote: $x = 6$ x-intercept: $x = 4$ y-intercept: $y = \ln 3$ 
(ii)	$\frac{dy}{dx} = \frac{-\frac{1}{2}}{3 - \frac{1}{2}x} = -\frac{1}{6 - x}$ When $x = 2$, $y = \ln 2$ and $\frac{dy}{dx} = -\frac{1}{4}$. Equation of tangent: $y - \ln 2 = -\frac{1}{4}(x - 2)$ $y + \frac{1}{4}x = \frac{1}{2} + \ln 2$ $4y + x = 2 + 4\ln 2$
(iii)	Area required $= \frac{1}{2}(\ln 2)(4\ln 2) - \int_2^4 \ln\left(3 - \frac{1}{2}x\right) dx$ $= 0.1883 \text{ (4 dec pl)}$

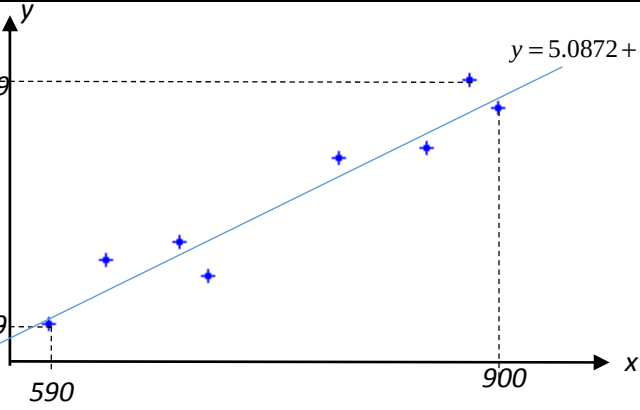
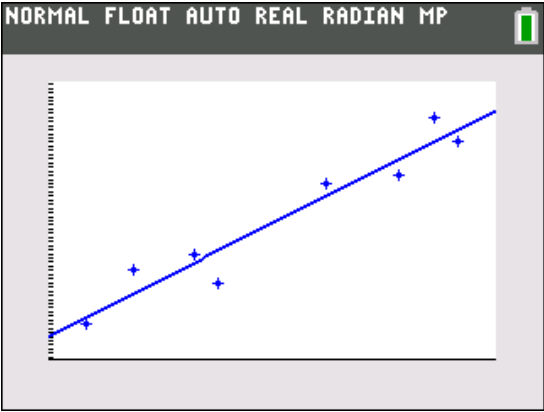
5 (i)	Let x, y and z be the amount of chemical samples of X, Y and Z in 200 ml of sample W respectively. $\frac{80}{200}x + \frac{120}{200}y + \frac{60}{200}z = 75$ $\frac{60}{200}x + \frac{60}{200}y + \frac{100}{200}z = 75$ $\frac{60}{200}x + \frac{20}{200}y + \frac{40}{200}z = 50$ Using GC, $x = 112.5, y = 12.5, z = 75$ There are 112.5 ml, 12.5 ml and 75 ml of samples X, Y and Z respectively.
(ii)	
(iii)	Maximum $P = 11.1$ It occurs when $t = 0.625$
(iv)	When $t = 3$, $\frac{dP}{dt} = -0.589$ The rate of change of potential energy is -0.589 kJ per second.
(v)	Required area $= \int_0^{0.5} 8 + 5e^{-0.4t} - e^{1-2t} dt$ $= \left[8t - \frac{25}{2}e^{-0.4t} + \frac{1}{2}e^{1-2t} \right]_0^{0.5}$ $= \left(4 - \frac{25}{2}e^{-0.2} + \frac{1}{2} \right) - \left(0 - \frac{25}{2} + \frac{1}{2}e \right)$ $= 17 - \frac{25}{2}e^{-0.2} - \frac{e}{2}$

6	Let T be random variable denoting mass of a cup of jelly. $T : N(\mu, \sigma^2)$ $P(T < 40) = 0.03$ $P\left(Z < \frac{40 - \mu}{\sigma}\right) = 0.03$ $\frac{40 - \mu}{\sigma} = -1.88079361$ $P(T > 50) = 0.08$ $P\left(Z \leq \frac{50 - \mu}{\sigma}\right) = 0.92$ $\frac{50 - \mu}{\sigma} = 1.405071561$ $\mu - 1.88079\sigma = 40 \dots\dots\dots (1)$ $\mu + 1.40507\sigma = 50 \dots\dots\dots (2)$ Solving (1) and (2), we get $\mu = 45.72389 = 45.7 \text{ (3 s.f.)}$ $\sigma = 3.04334 = 3.04 \text{ (3 s.f.)}$ $\text{Var}(T) = 3.04334^2 = 9.26$
7(i)	No. of codes = $5! \times 4!$ $= 2880$
(ii)	No. of codes = $4 \times 3 \times 6!$ $= 8640$
(iii)	Total number of codes = $8! = 40320$ No. of 'good' codes $= 4! \times 4! \times 2$ $= 1152$ Required probability = $\frac{1152}{40320} = \frac{1}{35}$ or 0.0286

8(i)	Tree diagram <u>First try</u> <u>Second try</u>
(ii)	$P(\text{Not win anything})$ $= 0.25 \times 0.7 + 0.75 \times 0.4 \times 0.7 + 0.75 \times 0.6$ $= 0.835$
(iii)	$P(2^{\text{nd}} \text{ Try} \mid \text{win a prize})$ $= \frac{0.75 \times 0.4 \times 0.3}{1 - 0.835}$ $= \frac{6}{11} \text{ or } ; 0.545$
(iv)	Let X be the r.v. denoting the number of games he won prize put of 5 games. $X \sim B(5, 0.165)$ $P(X \geq 3) = 1 - P(X \leq 2)$ $= 0.0345$

9(i)	The phrase 'large number' in the first sentence is required in order to assume that the probability of a faulty calculator is approximately constant at 0.05 with the event of getting a faulty calculator or not being independent.
(ii)	Let X be the r.v denoting the number of faulty calculators out of 30. $X \sim B(30, 0.05)$ $P(X > 3) = 1 - P(X \leq 3)$ $= 0.060772$ $= 0.0608 \text{ (3 s.f)}$
(iii)	Let Y be the r.v denoting the number of boxes that contains more than 3 faulty calculators out of 9. $Y \sim B(9, 0.060772)$ $P(Y = 1) = 0.331218$ Prob required $= 0.331218 (0.060772)$ $= 0.0201 \text{ (3 s.f)}$ <u>Mtd 2</u> Probability required $= \binom{9}{1} (0.060772)(1 - 0.060772)^8 \times 0.060772$ $= 0.0201$
	$E(X) = np = 30(0.05) = 1.5$ $\text{Var}(X) = npq = 1.5(0.95) = 1.425$ Since a sample size of 50 is large, Central Limit Theorem applies. $\bar{X} \sim N\left(1.5, \frac{1.425}{50}\right)$ approx. $P(\bar{X} < 1.6) = 0.723$

10(i)	Since the ornithologist wishes to test the bird watcher claim that whether the mean wing length of English male blackbirds is 131 mm is valid, he should use a 2-tail test.
(ii)	The ornithologist will be able to carry out the test without knowing the distribution of the wing lengths, because the sample size is relatively large ($n = 40$) and hence Central Limit Theorem can be applied, so that the sample mean wing length of English male blackbirds is normally distributed.
(iii)	$H_0 : \mu = 131$ $H_1 : \mu \neq 131$ Unbiased estimate of the population variance is $s^2 = \frac{40}{39} \times 11.84^2 = 143.7801026$ Assuming H_0 is true. Since sample size = 40 is large, by Central Limit Theorem, $\bar{X} \sim N\left(131, \frac{143.7801026}{40}\right)$ approximately. The test statistic is $Z = \frac{\bar{X} - 131}{\sqrt{143.7801026/40}} \sim N(0, 1)$ Using GC, $p\text{-value} = 0.0649$ (3 sig fig) This $p\text{-value}$ means that there is twice the probability of drawing a random sample of 40 English male blackbirds with sample mean wing lengths less than 127.5 mm, assuming that the population mean wing lengths of English male blackbirds is 131 mm.
(iv)	Let X be the r.v. denoting the wing length of a randomly chosen English male blackbird. $H_0 : \mu = 131$ $H_1 : \mu > 131$ Test at 5% significance level. Assuming H_0 is true. Since sample size = 40 is large, by Central Limit Theorem, $\bar{X} \sim N\left(131, \frac{150.7984}{40}\right)$ approximately. The test statistic is $Z = \frac{\bar{X} - 131}{\sqrt{150.7984/40}} \sim N(0, 1)$ Given conclusion: H_0 is rejected. Hence the value of the test statistic falls inside critical region. $\frac{k - 131}{\sqrt{150.7984/40}} \geq 1.64485 \quad \text{--- (*)}$ $k \geq 134.2 \quad (1 \text{ dec pl})$ The set of values for k required is $\{k \in \mathbb{R} : k \geq 134.2\}$

11 (i)	
(ii)	NORMAL FLOAT AUTO REAL RADIAN MP LinReg $y=a+bx$ $a=5.087188612$ $b=.1453990849$ $r^2=.9265627959$ $r=.9625813191$ $r = 0.9625813191$ $= 0.963$ (3 s.f.) This indicates very strong positive linear correlation between number of new subscribers and amount of profit generated. The larger the number of new subscribers, the more the amount of profit generated.
(iii)	$y = 5.0872 + 0.145399x$ $y = 5.087 + 0.145x$ 
(iv)	For every new subscriber, the online shopping site generates a profit of \$145. When there are no new subscriber, the online shopping site generates a profit of \$5087.

(v)	$100 = 5.0872 + 0.145399x$ $x = \frac{100 - 5.0872}{0.145399}$ $= 652.77$ $= 653 \text{ (nearest whole number)}$ The number of online shopping subscribers is 653. This estimate is reliable because the profit amount of \$100,000 is within data range of \$89,000 to \$139,000 and the product moment correlation coefficient of 0.963 is close to 1, indicating strong positive correlation between number of new subscribers and amount of profit generated
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12(i)	Let X g be the mass of a randomly chosen potato. $X \sim N(0.165, 0.015^2)$ Y g be the mass of a randomly chosen carrot. $Y \sim N(0.072, 0.012^2)$ Let $T = Y_1 + Y_2 - X$ $E(T) = 2(0.072) - 0.165 = -0.021$ $\text{Var}(T) = 2(0.012^2) + 0.015^2 = 0.000513$ $T \sim N(-0.021, 0.000513)$ Required probability $= P(-0.02 < T < 0.02)$ $= 0.447 \text{ (3 s.f.)}$
(ii)	Let A be the cost of a randomly chosen potato. $A = 2.5X$ $E(A) = 2.5(0.165) = 0.4125$ $\text{Var}(A) = 2.5^2(0.015^2) = 0.00140625$ $A \sim N(0.4125, 0.00140625)$ Required probability $= [P(A > 0.4)]^2 = 0.398 \text{ (3 s.f.)}$ Or use mass to find the answer. $P\left(X > \frac{0.4}{2.5}\right)$
(iii)	Part (iii) answer is greater since (ii) is a subset of the possible cases in (iii). For eg, the total cost of 2 potatoes can still be more than \$0.80 when one of them is less than \$0.40 (eg. \$0.30) and the other is more than \$0.40 (eg. \$0.60).
(iv)	Let W be the cost of 8 potatoes and 6 carrots. $\text{Let } W = 2.5(X_1 + X_2 + \dots + X_8) + 1.8(Y_1 + \dots + Y_4 + Y_5 + Y_6)$ $E(W) = 2.5(8 \times 0.165) + 1.8(6 \times 0.072)$ $= 4.0776$ $\text{Var}(W) = 2.5^2(8 \times 0.015^2) + 1.8^2(6 \times 0.012^2)$ $= 0.014049$ $W \sim N(4.0776, 0.014049)$ Required probability $= P(3.8 < W < 4.2) = 0.840 \text{ (3 s.f.)}$